



بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ
جامعة كرري - كلية علوم الحاسوب وتقانة المعلومات
المستوى الثالث (هبرمجيات)
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Section A: Definitions (20 marks)

Define the following terms clearly and concisely:

1. Buffer Overflow
2. Format String Vulnerability
3. SQL Injection
4. Race Condition
5. Mutex
6. Manual Code Auditing
7. Malware
8. Botnet
9. Vulnerability Assessment Tool
10. Semaphore

Section B: Multiple Choice Questions (10 marks)

Choose the correct answer from the options provided.

1. What type of injection attack uses `'; DROP TABLE users; --` to disrupt SQL queries?
A) XSS
B) SQL Injection
C) Format String
D) Buffer Overflow
2. Which synchronization method ensures only the owning thread can release a resource?
A) Semaphore
B) Monitor
C) Mutex
D) Deadlock
3. Which type of Trojan gives full remote control of a system?
A) General Trojan
B) Logic Bomb
C) RAT (Remote Access Trojan)
D) Payload Trojan

4. Which of the following is NOT a type of race condition result?
- A) Lost Update
 - B) Deadlock
 - C) Format Error
 - D) Livelock
5. What is the main function of a BotMaster?
- A) Repair infected systems
 - B) Analyze vulnerabilities
 - C) Control a network of Bots
 - D) Prevent malware infections
6. Which tool is mainly used for digital forensic analysis?
- A) Nessus
 - B) Cuckoo Sandbox
 - C) Autopsy
 - D) Bandit
7. Which vulnerability allows attackers to run arbitrary commands by exploiting formatting functions?
- A) SQL Injection
 - B) Format String
 - C) XSS
 - D) CSRF
8. A race condition usually occurs due to lack of:
- A) Compilation
 - B) Synchronization
 - C) Testing
 - D) Logging
9. What character is often used in SQL injection?
- A) ;
 - B) #
 - C) @
 - D) %
10. The most dangerous Trojan type is:
- A) Email Trojan
 - B) Remote-Access Trojan
 - C) General Trojan
 - D) Script Trojan

Section C: True or False (10 marks)

Indicate whether the following statements are True or False.

1. A Mutex can guard multiple resources.
2. Format string attacks involve injecting HTML code into web pages.
3. SQL Injection can lead to full control of the database server.
4. Worms require a host file to spread.
5. Manual code auditing is always faster than automated tools.
6. A race condition depends on the order of thread execution.
7. Malware can include viruses, worms, and spyware.
8. OpenVAS is a network traffic sniffer.
9. Monitors in Java are used for thread synchronization.
10. Buffer overflow can overwrite adjacent memory.

Section D: Enumerate (20 marks)

List the required items for each question.

1. List four steps in the manual code auditing process.
2. List three tools used for vulnerability assessment.
3. Enumerate three effects of SQL Injection attacks.
4. Mention three synchronization primitives discussed in the lectures.
5. List five types of software vulnerabilities.
6. Name two Java tools/plugins for secure coding.
7. List three types of malware.
8. Mention two examples of race condition outcomes.
9. List three best practices for using semaphores.
10. List three challenges in using security analysis tools.

Network security
network analysis
malware
vulnerabilities

Good luck